
Calibrating Mathematical Biases for Arctic Safety: A Hybrid XGBoost-NHITS Framework for Sea Ice Thickness Forecasting

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Abstract

The accelerated retreat of Arctic sea ice has markedly enhanced the economic viability of the Northern Sea Route. Accurate monitoring of sea ice thickness (SIT) is an essential prerequisite for safe navigation through these pathways. Nevertheless, satellite observations are constrained by the "Summer Observational Void", a persistent data gap from May to September arising from melt ponds that disrupt altimetry signals. Although numerical reanalysis models such as PIOMAS afford spatiotemporal continuity, they depend on mathematical simplifications and approximate thermodynamic principles, yielding systematic biases. These models frequently diverge substantially from observational data, commonly overestimating SIT by 4–8 m relative to actual conditions.

1. Backgrounds

The phenomenon of "Arctic Amplification", propelled by contemporary climate change, has accelerated Arctic sea ice decline, thereby paradoxically unveiling a "new sea of opportunity" for humankind. Notably, the Northern Sea Route—the most direct maritime corridor linking Northeast Asia and Europe—is ascending as a "third artery" in global logistics networks. By curtailing shipping distances by roughly 30–40% relative to conventional Suez Canal trajectories, the NSR harbors the capacity to profoundly reconfigure the international maritime trade paradigm.

These geoeconomic merits have spurred consistent year-on-year escalation in NSR cargo throughput, necessitating a reevaluation of its strategic prominence as a nexus for energy resource conveyance and mercantile navigation. Yet, beneath this vast economic promise lurks the perennial menace of capricious sea ice. Arctic SIT and configuration exhibit erratic fluctuations tied to seasonal and climatic vicissitudes, engendering acute perils of calamitous incidents, including vessel stranding or hull impairment.

Hence, for the NSR to actualize its role as a bona fide "commercial route", the institution of high-fidelity SIT prognostication technology is imperative. Beyond rudimentary binary ice discernment, such technology must gauge ice tenuousness to affirm safe transit for icebreakers and commercial tonnage. This investigation endeavors to prognosticate pan-Arctic SIT, thereby forecasting quotidian navigable corridors and furnishing "Actionable Intelligence" for maritime security.

2. Data and materials

This study adopts a Data-Centric approach, complementarily merging physical numerical models with satellite-based empirical data to precisely learn spatiotemporal patterns and forecast future Arctic SIT.

2.1 Physical prior: PIOMAS reanalysis data

PIOMAS data was utilized as the Physical Prior for sea ice thickness. While PIOMAS provides continuous estimates by assimilating ice concentration and velocity, it is inherently an estimate based on numerical modeling. Its mathematical idealizations lead to structural biases, such as underestimating localized variability or over-smoothing. This suggests that data calibrated with empirical observations possesses higher reliability in explaining actual sea ice thickness than raw physical model outputs.

2.2 Ground truth: NSIDC CryoSat-2 Level-4 data

The NASA National Snow and Ice Data Center (NSIDC) 'CryoSat-2 Level-4 Sea Ice Elevation, Freeboard, and Thickness (RDEFT4)' dataset was employed as the Ground Truth. This Level-4 product offers superior empirical precision compared to raw PIOMAS estimates. However, it is primarily valid from September to May, as summer melt ponds distort altimetry signals, resulting in the aforementioned "Summer Observational Void."

2.3 Construction of high-fidelity reconstructed dataset

Our strategy involved training an XGBoost model on the residuals between PIOMAS estimates and actual satellite observations during periods when NSIDC data is available (September–May). This calibration framework ensures that reconstructed values converge toward the high precision of NSIDC during observable periods, while inferring physically plausible values for the summer months and the uncertain 2025 period based on learned error patterns. The resulting High-Fidelity Reconstructed Dataset reflects the true Arctic environment more accurately than raw PIOMAS data, providing a robust foundation for 2026 forecasting.

2.4 Data preprocessing: spatial standardization

To integrate datasets with differing grid systems, a Spatial Regridding process was performed. We established the EASE-Grid 2.0 (25km) system as the reference grid. PIOMAS data was subsequently interpolated and mapped to this grid to ensure spatiotemporal consistency across the analytical pipeline.

3. Methodology

This study proposes a two-stage Physical-Informed Machine Learning Calibration framework to rectify physical model biases and generate reliable probabilistic forecasts. This framework leverages deep learning to address the unique challenges of Arctic sea ice dynamics, including the high variability and mobility of the ice pack (Landy et al., 2022). Specifically, the methodology incorporates uncertainty quantification to evaluate the dependability of predictions, thereby facilitating robust decision-making in safety-critical applications such as Arctic navigation (Ailuro et al., 2024).

3.1 Stage 1: Non-linear residual correction (XGBoost)

Machine learning is utilized as a non-linear residual learner to calibrate the systematic errors in numerical reanalysis. The XGBoost model learns the discrepancies between PIOMAS estimates and satellite observations to reconstruct high-fidelity SIT values during summer voids. This approach integrates data assimilation methods with deep learning, employing CryoSat-2 summer sea-ice thickness data derived from local variations in radar echo responses to improve forecasts despite challenges like roughness-induced electromagnetic range bias in heavily deformed ice (Song et al., 2024). This methodology enhances the assimilation of sea-ice concentration and thickness, which are crucial for improving both short-term and long-term sea-ice forecast skills (Song et al., 2024). This refinement is particularly valuable given that the accuracy of sea-ice forecasts is highly dependent on precise initial sea-ice states, and assimilating satellite observations is a common strategy to mitigate forecast errors (Song et al., 2024). Such assimilated data often needs careful consideration of potential biases, especially in reanalysis products where sea ice thickness might be indirectly estimated or subject to model-specific inaccuracies (Gao et al., 2021).

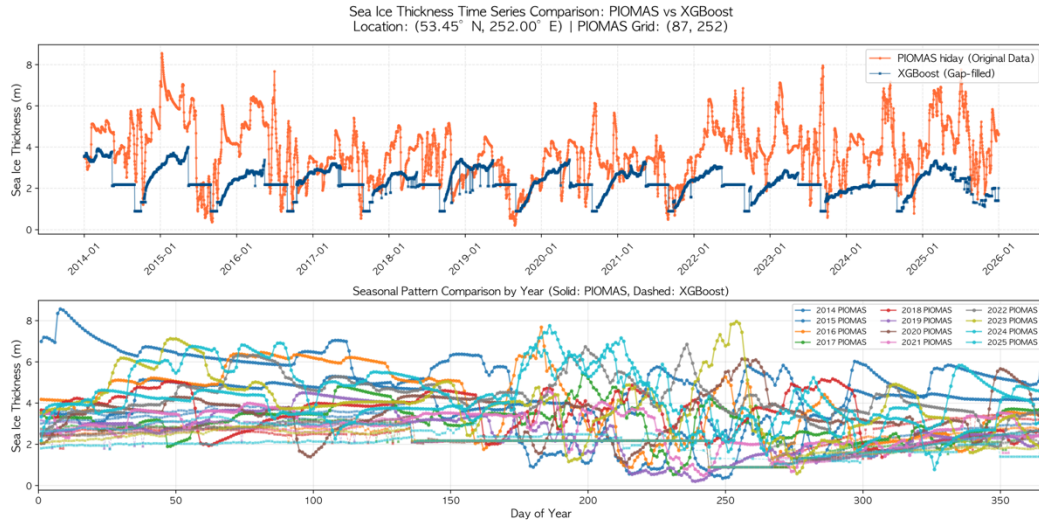


Figure1. Sea ice thickness time series comparison: PIOMAS v.s. XGBoost

Correction of Systematic Bias: Long-term time series show PIOMAS consistently overestimating thickness, while XGBoost rectifies these biases for realistic values.

Continuity in Summer Voids: XGBoost ensures time-series continuity by interpolating recurring "Summer Voids" in satellite records.

Stable Seasonal Cycles: XGBoost-calibrated results exhibit stable cycles, unlike PIOMAS's irregular fluctuations.

Detection of Thinning Trends: The model captures climate-driven thinning, evident in 2025 data versus prior years.

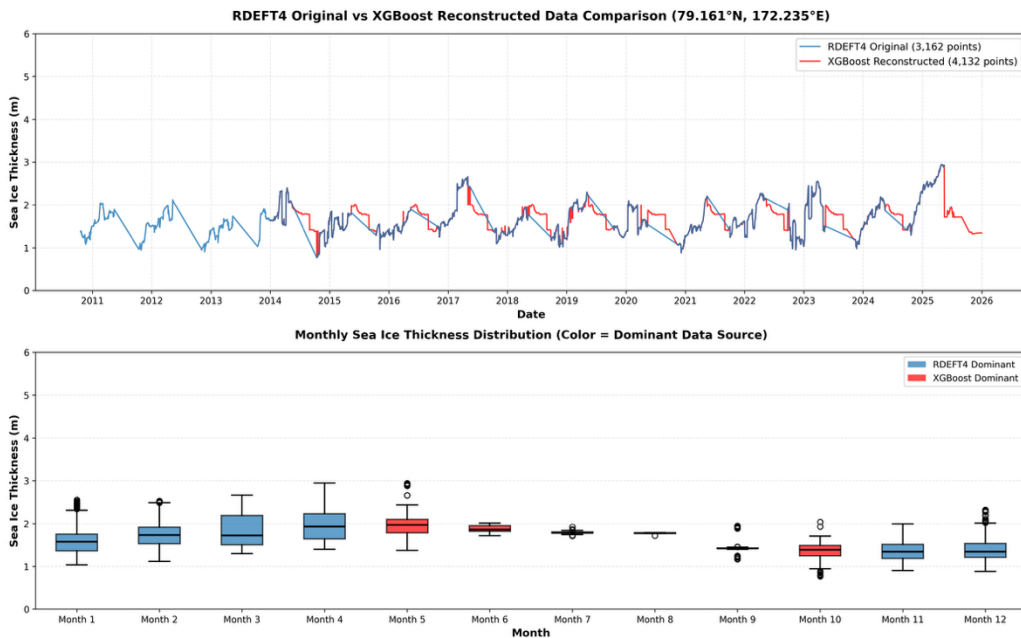


Figure 2. Time-series reconstruction results at a specific pixel (79.161N, 172.235E)

The blue solid line represents the original satellite observation data (RDEFT4), which clearly exhibits recurring "Summer Voids"—systematic seasonal missing values caused by the interference of melt ponds on altimetry signals. The red solid line indicates the output of the proposed XGBoost model, which successfully performs imputation of these missing intervals along physically plausible trajectories. The model demonstrates high fidelity to the ground truth during observable periods, validating the successful reconstruction of a continuous and robust sea ice thickness dataset.

3.2 Strategy for operational navigability assessment

Prior to analyzing the 2026 forecasts, we established a "Dual-Track Analysis Strategy" to translate the probabilistic predictions from the N-HITS model into actionable operational intelligence. Based on a conservative operational threshold ($\tau = 1.5m$), this strategy classifies the Arctic domain into risk and navigable zones to support strategic decision-making.

First, regarding the identification and exclusion of No-Go zones, pixels where the predicted sea ice thickness ($\hat{y}$) exceeds the operational limit (τ) are identified as "No-Go" zones ($\hat{y} > 1.5m$). Through numerical threshold filtering, these areas are designated as forbidden zones for navigation and are visualized with black masking. This effectively creates a "Safety Buffer," ensuring that high-risk areas are physically isolated from potential route planning to preclude collision risks.

Second, concerning the weighting for optimal routes within navigable zones, the framework assigns a "Resistance Weight" proportional to the ice thickness rather than treating them as binary "Go" states for pixels determined to be navigable ($\hat{y} \leq 1.5m$). This approach enables the identification of the "Least Resistance Path"—corridors with thinner ice within the safety zone—allowing vessels to maximize fuel efficiency and minimize icebreaking loads.

3.3 Stage 2: Hierarchical probabilistic forecasting (N-HITS)

The reconstructed time series is processed by the N-HITS model, which utilizes a hierarchical structure to disentangle long-term climate thinning trends from short-term seasonal volatility. Beyond deterministic point estimates, the model generates Prediction Intervals (PI) to quantify the inherent stochasticity of the Arctic environment. This probabilistic forecasting capability is crucial for accurately representing the complex, rapidly changing sea ice conditions, offering a more complete picture than traditional deterministic models (Gao et al., 2021). This approach moves beyond simple point forecasts to provide a more comprehensive understanding of future sea ice states, which is vital for assessing risks and opportunities in the Arctic region (Gao et al., 2021). Furthermore, these probabilistic forecasts offer valuable insights into the range of possible outcomes, enabling stakeholders to make more informed decisions regarding maritime operations and resource management in the Arctic (Ailuro et al., 2024; Gao et al., 2021). This is particularly salient for ice-strengthened vessels, where accurate sea ice thickness forecasts directly influence navigational performance and safety (Min et al., 2025). The ability to account for spatial correlation in thickness forecasts further enhances the utility of these probabilistic forecasts, as demonstrated by their closer alignment with nominal empirical coverage rates compared to models that do not consider spatial dependencies (Gao et al., 2021).

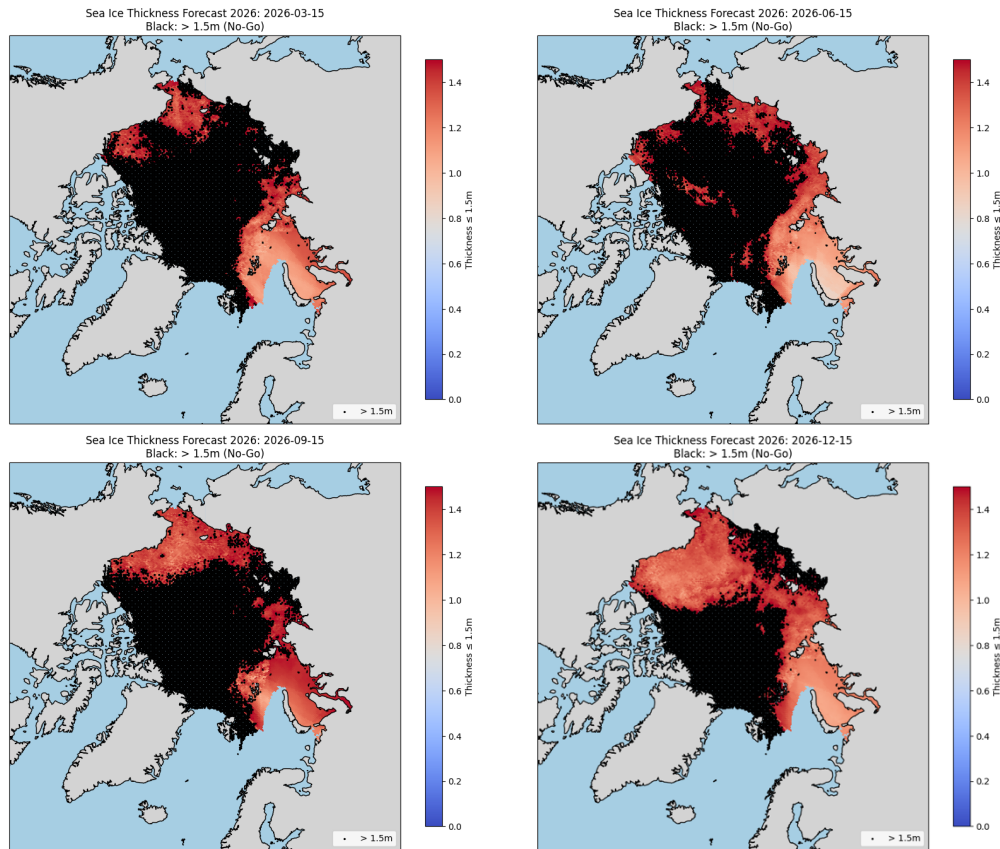


Figure 3. Spatiotemporal probabilistic SIT Forecasts for 2026 via N-HITS

This figure illustrates the quarterly evolution of Arctic SIT throughout 2026, highlighting the model's ability to capture the high-frequency seasonal transitions characteristic of the polar environment. Black dots represent "No-Go Zones," where SIT exceeds the 1.5m safety threshold for standard commercial vessels.

Q1 (March 15): Peak freezing and maximum navigation constraints

The forecast for mid-March shows the maximum extent of hazardous ice coverage across the Arctic basin.

The central Arctic and the majority of the NSR are dominated by "No-Go Zones" exceeding 1.5m in thickness, necessitating heavy icebreaker support for any maritime movement.

Q2 (June 15): Melt onset and marginal thinning

By mid-June, the transition toward the melt season becomes evident as SIT values at the ice margins begin to decrease.

While the central Arctic remains largely impenetrable, the model captures a gradual reduction in thickness along the Siberian coast, reflecting the early stages of seasonal thermal forcing.

Q3 (September 15): Annual minimum and navigable path identification

The September forecast represents the annual SIT minimum, showing a dramatic contraction of the "No-Go Zone".

Significant navigable corridors appear along the Russian coast, with thickness values dropping into the 0.5m–1.2m range, providing the widest window for commercial transit without specialized icebreaking assistance.

Q4 (December 15): Re-freezing and expansion of hazardous zones

1 As the Arctic enters the winter cycle, the N-HITS model accurately simulates the rapid re-freezing
2 process.
3 The "No-Go" threshold ($> 1.5\text{m}$) rapidly re-establishes itself across the central basin,
4 demonstrating the model's sensitivity to multi-rate temporal dynamics during the freezing
5 transition.
6

7 4. Results

8 4.1 Performance in data restoration and reconstruction

9 The Stage 1 restoration model achieved an R^2 of 0.9207 and an RMSE of 0.1734m demonstrating
10 high precision in data reconstruction. Long-term comparison (2014–2026) confirms that the 4m to 8m
11 overestimation bias in PIOMAS was successfully calibrated to a more realistic 2m to 4m range. This
12 substantial reduction in bias highlights the effectiveness of the machine learning residual correction in
13 aligning numerical reanalysis outputs with observed sea ice thickness (Gregory et al., 2025). The
14 calibrated data thus provides a more accurate representation of sea ice thickness, addressing a
15 significant limitation of relying solely on physical models (Gao et al., 2021). This improved data
16 serves as a more reliable foundation for subsequent probabilistic forecasting, enhancing the overall
17 accuracy and utility of the predictions. The approach effectively mitigates the "Summer Observational
18 Void" by generating robust sea ice thickness estimates during periods of limited satellite data, thus
19 filling critical gaps in monitoring capabilities. This capability is especially pertinent given the rapid
20 decrease in Arctic sea ice thickness observed over recent decades, underscoring the urgent need for
21 consistent and accurate monitoring (Bocquet et al., 2024). Furthermore, the accuracy of sea ice
22 thickness measurements is vital for understanding its role in climate regulation, supporting marine
23 biodiversity, and influencing biogeochemical cycles such as nutrient cycling and carbon sequestration
24 (Bilgili et al., 2025).

25 4.2 Probabilistic Forecasting Validation (May 2025)

26 Final model validation for the May 2025 period yielded the following results:

27 Accuracy: The model recorded an RMSE of 0.1965m and a MAE of 0.1186m

28 Correlation and Bias: An Anomaly Correlation Coefficient (ACC) of 0.8304 was achieved, with a
29 minimal Bias of -0.0344m

30 Reliability: The Prediction Interval Coverage Probability (PICP) reached 0.9073, indicating that over
31 90% of actual values were captured within the predicted confidence buffer.

32 Operational Insight: Under an average SIT of 2.015m, the model identified a navigable area ratio of
33 14.00.

34

35 **Table 1. Statistical evaluation of model accuracy and predictive reliability.**

Stage	Model	Metric	Value
Stage 1: Restoration	XGBoost	R^2	0.9207
		RMSE	0.1734 m
		RMSE	0.2012 m
Stage 2: Forecasting	N-HITS	ACC	0.8489
		PICP	0.9742

36

37 5. Discussion

38 This hybrid framework effectively bridges the gap between physical modeling constraints and
39 empirical reality. The ability of the N-HITS model to capture the distinct thinning trend in 2025

1 compared to 2024 reflects its sensitivity to accelerating Arctic warming. Although the Skill Score of -
2 0.0175 indicates challenges in surpassing persistence models during volatile transition months, the
3 high PICP of 0.9073 provides a robust "Safety Margin" for maritime risk management. However, a
4 counterargument posits that hybrid machine learning-physical models, while promising, exhibit
5 limitations such as fluctuating RMSE as lead times increase, particularly in unstable conditions,
6 potentially undermining reliability in long-term forecasts (Sureja, 2024). This demonstrates the
7 framework's potential in providing actionable intelligence for Arctic operations, even amidst rapid
8 environmental changes, by quantifying uncertainty. The integration of machine learning techniques
9 with physical models offers a robust approach to sea ice forecasting, overcoming limitations inherent
10 in standalone methods (Sureja, 2024).

11 **6. Limitations and Study Scope**

12 To ensure analytical precision and operational validity, this study employed two specific criteria for
13 data selection and navigability assessment.

14 First, the analysis was restricted exclusively to target pixels exhibiting a sea ice concentration of 70%
15 or higher. This threshold was adopted to concentrate on the substantial pack ice zone while mitigating
16 noise arising from the highly variable marginal ice zone. Although this approach effectively minimizes
17 satellite observation errors associated with low-density ice, it necessarily precludes detailed
18 examination of regions with SIC below 70%.

19 Second, with respect to navigational safety, regions with sea ice thickness exceeding 1.5 m were
20 designated as "No-Go Zones." This criterion constitutes a conservative operational presumption,
21 aligned with the typical continuous icebreaking capabilities of medium-class icebreakers.
22 Nevertheless, actual navigability is subject to variation based on specific vessel characteristics and
23 local ice deformation phenomena, such as pressure ridges.

25 **7. Conclusion**

26 By integrating bias-corrected physical reanalysis with hierarchical probabilistic machine learning, this
27 framework delivers actionable intelligence for Arctic operations. Its strong performance in data
28 restoration and probabilistic forecasting addresses the summer observational void, providing a reliable
29 tool for improving navigation safety and strategic planning along the Northern Sea Route. Future
30 research could investigate causal inference methods, such as multivariate Granger causality and
31 PCMCi+, to identify direct drivers of sea ice dynamics, transitioning from correlation- to causation-
32 based modeling to enhance robustness, interpretability, and predictive accuracy (Hossain et al., 2025;
33 Hossain & Gani, 2025). Moreover, generative AI models like GenSIM could improve Arctic sea ice
34 state predictions via high-fidelity, sub-daily forecasts (12-hour increments) incorporating complex ice-
35 ocean interactions without explicit physical modeling (Finn et al., 2025). These methods may mitigate
36 the computational intensity of traditional causal graph techniques like PCMCi+ and MVGC (Hossain
37 et al., 2025), while hybrid model ensembles tailored to lead times—e.g., DLPCMCI+ for short-term
38 (1-month), DLGC for intermediate, and DLDPCMCI+ for extended (5–6 months) forecasts—could
39 further improve performance, as no single model consistently outperforms others across horizons
40 (Hossain et al., 2025).

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